

Evaluate limits using Maclaurin's series

Key Points

To find limit of

$$\lim_{x \rightarrow a} \frac{N(x)}{D(x)}$$

Problems with limits arise from

$$\text{" } \frac{0}{0}, \frac{\infty}{\infty}, \frac{\infty}{0} \text{"}$$

As $x \rightarrow 0$ we divide by the smallest power of x . As $x \rightarrow \infty$ we divide by the largest power of x

To use Maclaurin's series we replace non-polynomial functions with their Maclaurin's series before finding the limit

Basic Skills

Q1. Find the Maclaurin's series for

a) $x \sin x$ b) $\cos^2 x$ c) $e^{5x} \sin x$

d) $(3-x)^{-1}$ d) $\ln^2(1+x)$

Q2. What happens to the graph of $f(x) = \frac{5x^3+1}{x+1}$ as $x \rightarrow \infty$

Q3. State the value of the following limits

a) $\lim_{x \rightarrow \infty} \frac{x^2+5x}{2x^2-6x-2}$

b) $\lim_{x \rightarrow 0} \frac{x^2+5x}{2x^2-6x-2}$

c) $\lim_{x \rightarrow \infty} \frac{2x^3-2}{x^6-2}$

d) $\lim_{x \rightarrow 0} \frac{2x^3-2}{x^6-2}$

e) $\lim_{x \rightarrow 0} \frac{5x+\sqrt{x}}{x+1}$

f) $\lim_{x \rightarrow \infty} \frac{(x^5+1)^2}{x^6+5x-1}$

Competency Skills

Q4. State the limit of the function $f(x) = \frac{(x-4)^2}{x^2+4}$ as $x \rightarrow 4$

Q5. By replacing $\cos x$ with it's Maclaurin's series find $\lim_{x \rightarrow 0} \frac{\cos x - 1}{5x^2}$

Q6. By replacing e^{4x} with it's Maclaurin's series find $\lim_{x \rightarrow 0} \frac{e^{4x} - (1+4x)}{5x^2}$

Q7. Find the following limits

a) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$

b) $\lim_{x \rightarrow 0} \frac{x^2+2}{e^x}$

c) $\lim_{x \rightarrow \infty} \frac{x^2+2}{e^x}$

d) $\lim_{x \rightarrow 0} \frac{\sqrt{2-x}+4}{x}$

e) $\lim_{x \rightarrow \infty} \frac{\ln(1+x)}{2x+x^2}$

f) $\lim_{x \rightarrow 0} \frac{(x+5) \sin x}{\cos x}$

g) $\lim_{x \rightarrow 0} \frac{2-e^x \sin x}{\cos\left(x+\frac{\pi}{2}\right)}$

Advanced Skills

Q8. Explain when we can just use direct substitution of the limit into the function

Q9. Find the limit $\lim_{x \rightarrow \infty} f(x)$ where $f(x) = \frac{e^x+e^{-x}}{e^x-e^{-x}}$ (i.e. the function is hyperbolic cot)

Q10. Find the limit $\lim_{x \rightarrow 0} \frac{\ln(1+\sin 2x)-x}{e^{5x}-1-5x}$

Q11. a) Find the first 3 non-zero terms of the Maclaurin's series for the function $f(x) = \ln(\cos x + \sin x)$

b) Is it possible to evaluate the limit $\lim_{x \rightarrow \infty} \frac{\ln(\cos x + \sin x)}{5x^2-9}$

Extension

Q12. Try to predict the limit in Q11, is it what you expected?

Hint: Try some values/plotting the graph